WMDR2-core

Package in package 'Model'

WMDI
Version 0.1 Phase 1.0 Pr
jkl created on 2026-03-06. Last modified 202

WMDR2-core diagram

Class diagram in package 'WMDR2-core'

WMDR2core
Notes
1. temporalExtent shall be realized as time:interval
2. temporal<Name> elements other than temporalExtent shall be realized as an array of two arrays, one for times, the values of <Name>, e.g., temporalTerritory = [[2000-01-01, 2024-07-13], ["https://codes.wmo.int/wmdr/territory/<ISO1>", "https://codes.wmo.int/wmdr/territory/<ISO2>"]].
3. Following the OGC Features model, geometry = geolocation = coordinates + elevation

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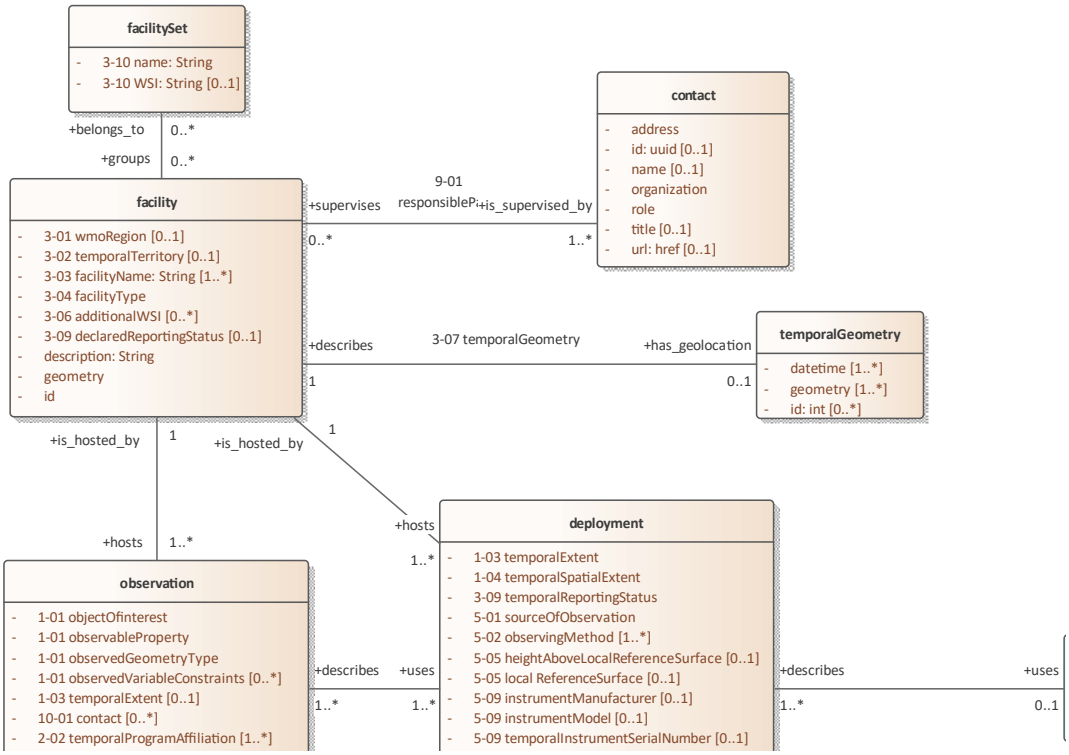

Figure 1: WMDR2-core

contact

Class in package 'WMDR2-core'

impkement according to VCard 4.0 specification [0].

ATTRIBUTES

address : Private

ADR: The structured type value consists of a sequence of
address components. The component values MUST be specified in
their corresponding position. The structured type value
corresponds, in sequence, to
the post office box;
the extended address (e.g., apartment or suite number);
the street address;
the locality (e.g., city);
the region (e.g., state or province);
the postal code;
the country name (full name in the language specified in
Section 5.1).

Example:
ADR;TYPE=work:;;Federal Department of Home Affairs FDHA Federal Office of M
eteorology and Climatology MeteoSwiss Measurements and Data\nOperation Cen
ter 1 (OPC 1)\nP.O. Box 257;Zürich-Flughafen;;CH-8058;Switzerland

[Is static True. Containment is Not Specifi

id : uuid Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specifi

name : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Either a full name, or a sequence of name parts. Examples:

Dr. Jörg Klausen (this corresponds to FN in vcard)
Klausen;Jörg;Dr.; (this corresponds to N in vcard)

[Is static True. Containment is Not Specifi

organization : Private

Example:
ORG:Federal Office of Meteorology and Climatology MeteoSwiss

[Is static True. Containment is Not Specifi

role : Private

[Is static True. Containment is Not Specifi

title : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specifi

ATTRIBUTES

ASSOCIATIONS

Association (direction: Unspecified) 9-01 responsibleParty

Source: Public supervises (Class) facility
Cardinality: [0..*]

Target: Public is_supervised_by (Class) contact
Cardinality: [1..*]

deployment

Class in package 'WMDR2-core'

"deployment" (or "Deployment") refers to the act of placing an instrument (or human observer) at a location to col measurements or take samples to be analyzed. In many cases, an instrument (or sensor package) can serve differen observations.

Deployments can exist in parallel or in series, one after the other.

The intent is to create a new deployment whenever the observingMethod changes over time.

The same deployment may be used if observations are made with the same brand/model of instrument, even if the specific instrument is rotated in and out, for example as part of operational maintenance cycles in a network.

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ATTRIBUTES

1-03 temporalExtent : Private
Alias: records:interval

[Is static True. Containment is Not Specifi

1-04 temporalSpatialExtent : Private
Alias: records:geometry

Typical spatial geo-referenced volume covered by the observations.

EXAMPLES:
(a) Air temperature at a surface observing site: Sydney Airport, New South Wales, Australia: lat. 33.9465°N, lon. 151.1731 alt. 6.0 m above mean sea level (amsl);
(b) The projected area or volume of the cone around a particular weather radar with a maximum range of 370 km (radar reflectivity) and 150 km (Doppler); to be expressed as a geometric shape;
(c) Three-dimensional grid of radar pixels;
(d) For infrared and visible imagery by meteorological satellite (sun-synchronous): global;
(e) For nadir sounding: atmospheric column above ocean;
(f) The Japan Meteorological Agency Advanced Himawari Imager has a 10 min cycle full disk scan (0.5-2.0 km spatial resolution) located above 140.68°E geostationary orbit at an altitude of 35,786 km;
(g) River discharge by gauge: size and geometric shape of a river catchment.

ATTRIBUTES

3-09 temporalReportingStatus : Private

[Is static True. Containment is Not Specifi

5-01 sourceOfObservation : Private

[Is static True. Containment is Not Specifi

5-02 observingMethod : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

Realized as an array of dictionaries, i.e. key-value pairs:
key: observingMethod id
values: observation id

[Is static True. Containment is Not Specifi

5-05 heightAboveLocalReferenceSurface : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

implement as array

[Is static True. Containment is Not Specifi

5-05 local ReferenceSurface : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Linked to heightAboveReferenceSurface

[Is static True. Containment is Not Specifi

5-09 instrumentManufacturer : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Change of instrument requires a new deployment

[Is static True. Containment is Not Specifi

5-09 instrumentModel : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Change of instrument requires a new deployment

[Is static True. Containment is Not Specifi

5-09 temporalInstrumentSerialNumber : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specifi

5-12 temporalGeometry : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

- Geospatial location of instrument/sensor

[Is static True. Containment is Not Specifi

ATTRIBUTES

indicates the end (hour) of a 24-hour period. Mandatory for 24-hour periodic observations, e.g., cumulative amount of precipitation

[Is static True. Containment is Not Specifi

6-08 temporalObservingSchedule : Private

complex type: when are observations made? Should be expressed as a temporal coverage, e.g. "from Feb-Oct on working d from 06-18 UTC"

[Is static True. Containment is Not Specifi

9-02 temporalDataPolicy : Private

remove from WMDR2 and leave for WCMP2?

[Is static True. Containment is Not Specifi

id : Private

[Is static True. Containment is Not Specifi

temporalOfficialStatus : Private

[Is static True. Containment is Not Specifi

ASSOCIATIONS

Association (direction: Unspecified)

Source: Public describes (Class) deployment
Cardinality: [1..*]

Target: Public uses (Class) procedure
Cardinality: [0..1]

ObservingProcedure(s)

Association (direction: Unspecified) 7-03 temporalReportingSchedule

Source: Public describes (Class) deployment
Cardinality: [1..*]

Target: Public reports_with (Class) temporalReportingSchedule
Cardinality: [1]

Association (direction: Unspecified) 5-09 instrument

Source: Public describes (Class) deployment
Cardinality: [0..*]

Target: Public uses (Class) instrument
Cardinality: [0..1]

Association (direction: Unspecified)

Source: Public describes (Class) observation
Cardinality: [1..*]

Target: Public uses (Class) deployment
Cardinality: [1..*]

ASSOCIATIONS

facility

Class in package 'WMDR2-core'

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ATTRIBUTES

3-01 wmoRegion : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

mandatory for fixed facilities, optional for mobile facilities.

[Is static True. Containment is Not Specifi

3-02 temporalTerritory : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

mandatory for fixed facilities, optional for mobile facilities

[Is static True. Containment is Not Specifi

3-03 facilityName : String Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specifi

3-04 facilityType : Private

[Is static True. Containment is Not Specifi

3-06 additionalWSI : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

An array of well-formed WSIs

[Is static True. Containment is Not Specifi

3-09 declaredReportingStatus : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Condition: must be provided if no deployment is specified; otherwise: Optional, inferred as "best case" from deployments: temporalReportingStatus + temporalProgramAffiliation.
Exception: "closed" over-rules deployment, in which case all deployments should be terminated.

[Is static True. Containment is Not Specifi

description : String Private

[Is static True. Containment is Not Specifi

ATTRIBUTES

geometry = geolocation = coordinates + elevation

[Is static True. Containment is Not Specifi

id : Private
Alias: 3-06 primaryWSI

[Is static True. Containment is Not Specifi

ASSOCIATIONS

Association (direction: Unspecified)

Source: Public is_hosted_by (Class) facility
Cardinality: [1]

Target: Public hosts (Class) observation
Cardinality: [1..*]

Association (direction: Unspecified) 3-07 temporalGeometry

Source: Public describes (Class) facility
Cardinality: [1]

Target: Public has_geolocation (Class) temporalGeometry
Cardinality: [0..1]

Association (direction: Unspecified) 9-01 responsibleParty

Source: Public supervises (Class) facility
Cardinality: [0..*]

Target: Public is_supervised_by (Class) contact
Cardinality: [1..*]

Association (direction: Unspecified)

Source: Public (Class) facility

Target: Public (InformationItem) temporalGeometry

Association (direction: Unspecified)

Source: Public groups (Class) facility
Cardinality: [0..*]

Target: Public belongs_to (Class) facilitySet
Cardinality: [0..*]

Association (direction: Unspecified)

Source: Public is_hosted_by (Class) facility
Cardinality: [1]

Target: Public hosts (Class) deployment
Cardinality: [1..*]

facilitySet

Class in package 'WMDR2-core'

ATTRIBUTES

3-10 name : String Private

[Is static True. Containment is Not Specifi

3-10 WSI : String Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Can facilitySet have a WSI? If not, this can be dropped.

[Is static True. Containment is Not Specifi

ASSOCIATIONS

Association (direction: Unspecified)

Source: Public groups (Class) facility
Cardinality: [0..*]

Target: Public belongs_to (Class) facilitySet
Cardinality: [0..*]

instrument

Class in package 'WMDR2-core'

inst
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ATTRIBUTES

id : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specifi

manufacturer : Private

This should come from a registry/catalogue, but should accept any organization.

[Is static True. Containment is Not Specifi

model : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

This should come from a registry/catalogue, but can accept "custom-built".

[Is static True. Containment is Not Specifi

ASSOCIATIONS

Association (direction: Unspecified) 5-09 instrument

observation

Class in package 'WMDR2-core'

observation (or observingCapability) wraps one or several deployments used to generate time series of observation the same observed variable. Observed variables can be simple or more complex, so they are represented as a comp of 4 aspects. Example are as follows:

- observableProperty: temperature, pressure, thickness of sea ice, concentration of CO2, scattering coefficient of a particles, ...
- observablePropertyConstraints: 2m above ground, PM2.5, 40% relative humidity, 635 nm, ...
- observedGeometryType: point, total column, profile
- objectOfInterest: atmosphere, interstitial air in the root space, air in an ice core, sea surface, soil

These entities will be supported by code lists from codes.wmo.int/wmdr2.

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ATTRIBUTES

1-01 objectOfinterest : Private

observedVariable should be modeled using the I-ADOPT ontology approach. Combination of
- observableProperty: a state variable (e.g., temperature), or another property (e.g., concentration of something)
- objectOfInterest: e.g., dry air (of the atmosphere)
- observedVariableConstraint(s): e.g., a size fraction, a specific wavelength, a pressure (range), a temperature (range), a humidity (range), etc., as relevant for the result of the observation. Constraints shall be documented as name-value pairs, where the names shall come from a code list, and the values shall come from a code list depending on the name, or be free text.

[Is static True. Containment is Not Specifi

1-01 observableProperty : Private

observedVariable should be modeled using the I-ADOPT ontology approach. Combination of
- observableProperty: a state variable (e.g., temperature), or another property (e.g., concentration of something)
- objectOfInterest: e.g., dry air (of the atmosphere)
- observedVariableConstraint(s): e.g., a size fraction, a specific wavelength, a pressure (range), a temperature (range), a humidity (range), etc., as relevant for the result of the observation. Constraints shall be documented as name-value pairs, where the names shall come from a code list, and the values shall come from a code list depending on the name, or be free text.

[Is static True. Containment is Not Specifi

1-01 observedGeometryType : Private

[Is static True. Containment is Not Specifi

1-01 observedVariableConstraints : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

observedVariable should be modeled using the I-ADOPT ontology approach. Combination of
- observableProperty: a state variable (e.g., temperature), or another property (e.g., concentration of something)
- objectOfInterest: e.g., dry air (of the atmosphere)
- observedVariableConstraint(s): e.g., a size fraction, a specific wavelength, a pressure (range), a temperature (range), a

ATTRIBUTES

1-03 temporalExtent : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

aggregate derived from deployments if deployments are provided; otherwise, specify begin-end

[Is static True. Containment is Not Specifi

10-01 contact : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

ogc:contact includes "role"

[Is static True. Containment is Not Specifi

2-02 temporalProgramAffiliation : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specifi

7-14 internationalExchange : Boolean Private

[Is static True. Containment is Not Specifi

7-14 internationalReportingSchedule : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

complex type: when are observations reported internationally? Should be expressed as a temporal coverage

Conditional: report if internationalExchange==True

[Is static True. Containment is Not Specifi

id : Private

[Is static True. Containment is Not Specifi

ASSOCIATIONS

Association (direction: Unspecified)

Source: Public describes (Class) observation
Cardinality: [1..*]

Target: Public uses (Class) deployment
Cardinality: [1..*]

Association (direction: Unspecified)

Source: Public is_hosted_by (Class) facility
Cardinality: [1]

Target: Public hosts (Class) observation
Cardinality: [1..*]

procedure

- ...

pro
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ASSOCIATIONS

Association (direction: Unspecified)

Source: Public describes (Class) deployment
Cardinality: [1..*]

Target: Public uses (Class) procedure
Cardinality: [0..1]

ObservingProcedure(s)

temporalGeometry

Class in package 'WMDR2-core'

temporalGe
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ATTRIBUTES

datetime : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)
[Is static True. Containment is Not Specifi

geometry : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)
[Is static True. Containment is Not Specifi

id : int Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)
[Is static True. Containment is Not Specifi

ASSOCIATIONS

Association (direction: Unspecified) 3-07 temporalGeometry

Source: Public describes (Class) facility
Cardinality: [1]

Target: Public has_geolocation (Class) temporalGeometry
Cardinality: [0..1]

temporalReportingSchedule

ATTRIBUTES

id : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
[Is static True. Containment is Not Specifi

interval : Private

An array of arrays of begin(inclusive) and end(exclusive) datetime stamps. If the end is not specified, this indicates open-ended. Intervals can overlap to indicate multiple schedules operating in parallel.
[Is static True. Containment is Not Specifi

schedule : Private

An array of arrays, whereby each nested array corresponds to an interval and describes a schedule. Schedules are specified follows:

...
[Is static True. Containment is Not Specifi

ASSOCIATIONS

Association (direction: Unspecified) 7-03 temporalReportingSchedule

Source: Public describes (Class) deployment
Cardinality: [1..*]

Target: Public reports_with (Class) temporalReportingSchedule
Cardinality: [1]

temporalGeometry

InformationItem in package 'WMDR2-core'

temporalGe
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ASSOCIATIONS

Association (direction: Unspecified)

Source: Public (Class) facility

Target: Public (InformationItem) temporalGeometry